

3.B The General Künneth Formula

Künneth formulas describe the homology or cohomology of a product space in terms of the homology or cohomology of the factors. In nice cases these formulas take the form $H_*(X \times Y; R) \approx H_*(X; R) \otimes H_*(Y; R)$ or $H^*(X \times Y; R) \approx H^*(X; R) \otimes H^*(Y; R)$ for a coefficient ring R . For the case of cohomology, such a formula was given in Theorem 3.16, with hypotheses of finite generation and freeness on the cohomology of one factor. To obtain a completely general formula without these hypotheses it turns out that homology is more natural than cohomology, and the main aim in this section is to derive the general Künneth formula for homology. The new feature of the general case is that an extra Tor term is needed to describe the full homology of a product.

The Cross Product in Homology

A major component of the Künneth formula is a **cross product** map

$$H_i(X; R) \times H_j(Y; R) \xrightarrow{\times} H_{i+j}(X \times Y; R)$$

There are two ways to define this. One is a direct definition for singular homology, involving explicit simplicial formulas. More enlightening, however, is the definition in terms of cellular homology. This necessitates assuming X and Y are CW complexes, but this hypothesis can later be removed by the technique of CW approximation in §4.1. We shall focus therefore on the cellular definition, leaving the simplicial definition to later in this section for those who are curious to see how it goes.

The key ingredient in the definition of the cellular cross product will be the fact that the cellular boundary map satisfies $d(e^i \times e^j) = de^i \times e^j + (-1)^i e^i \times de^j$. Implicit in the right side of this formula is the convention of treating the symbol $\times$ as a bilinear operation on cellular chains. With this convention we can then say more generally that $d(a \times b) = da \times b + (-1)^i a \times db$ whenever a is a cellular i -chain and b is a cellular j -chain. From this formula it is obvious that the cross product of two cycles is a cycle. Also, the product of a boundary and a cycle is a boundary since $da \times b = d(a \times b)$ if $db = 0$, and similarly $a \times db = (-1)^i d(a \times b)$ if $da = 0$. Hence there is an induced bilinear map $H_i(X; R) \times H_j(Y; R) \rightarrow H_{i+j}(X \times Y; R)$, which is by definition the cross product in cellular homology. Since it is bilinear, it could also be viewed as a homomorphism $H_i(X; R) \otimes_R H_j(Y; R) \rightarrow H_{i+j}(X \times Y; R)$. In either form, this cross product turns out to be independent of the cell structures on X and Y .

Our task then is to express the boundary maps in the cellular chain complex $C_*(X \times Y)$ for $X \times Y$ in terms of the boundary maps in the cellular chain complexes $C_*(X)$ and $C_*(Y)$. For simplicity we consider homology with $\mathbb{Z}$ coefficients here, but the same formula for arbitrary coefficients follows immediately from this special case. With $\mathbb{Z}$ coefficients, the cellular chain group $C_i(X)$ is free with basis the i -cells of X , but there is a sign ambiguity for the basis element corresponding to each cell e^i ,

namely the choice of a generator for the $\mathbb{Z}$ summand of $H_i(X^i, X^{i-1})$ corresponding to e^i . Only when $i = 0$ is this choice canonical. We refer to these choices as ‘choosing orientations for the cells.’ A choice of such orientations allows cellular i -chains to be written unambiguously as linear combinations of i -cells.

The formula $d(e^i \times e^j) = de^i \times e^j + (-1)^i e^i \times de^j$ is not completely canonical since it contains the sign $(-1)^i$ but not $(-1)^j$. Evidently there is some distinction being made between the two factors of $e^i \times e^j$. Since the signs arise from orientations, we need to make explicit how an orientation of cells e^i and e^j determines an orientation of $e^i \times e^j$. Via characteristic maps, orientations can be obtained from orientations of the domain disks of the characteristic maps. It will be convenient to choose these domains to be cubes since the product of two cubes is again a cube. Thus for a cell e_α^i we take a characteristic map $\Phi_\alpha: I^i \rightarrow X$ where I^i is the product of i intervals $[0, 1]$. An orientation of I^i is a generator of $H_i(I^i, \partial I^i)$, and the image of this generator under $\Phi_{\alpha*}$ gives an orientation of e_α^i . We can identify $H_i(I^i, \partial I^i)$ with $H_i(I^i, I^i - \{x\})$ for any point x in the interior of I^i , and then an orientation is determined by a linear embedding $\Delta^i \rightarrow I^i$ with x chosen in the interior of the image of this embedding. The embedding is determined by its sequence of vertices $v_0, \dots, v_i$. The vectors $v_1 - v_0, \dots, v_i - v_0$ are linearly independent in I^i , thought of as the unit cube in $\mathbb{R}^i$, so an orientation in our sense is equivalent to an orientation in the sense of linear algebra, that is, an equivalence class of ordered bases, two ordered bases being equivalent if they differ by a linear transformation of positive determinant. (An ordered basis can be continuously deformed to an orthonormal basis, by the Gram-Schmidt process, and two orthonormal bases are related either by a rotation or a rotation followed by a reflection, according to the sign of the determinant of the transformation taking one to the other.)

With this in mind, we adopt the convention that an orientation of $I^i \times I^j = I^{i+j}$ is obtained by choosing an ordered basis consisting of an ordered basis for I^i followed by an ordered basis for I^j . Notice that reversing the orientation for either I^i or I^j then reverses the orientation for I^{i+j} , so all that really matters is the order of the two factors of $I^i \times I^j$.

Proposition 3B.1. *The boundary map in the cellular chain complex $C_*(X \times Y)$ is determined by the boundary maps in the cellular chain complexes $C_*(X)$ and $C_*(Y)$ via the formula $d(e^i \times e^j) = de^i \times e^j + (-1)^i e^i \times de^j$.*

Proof: Let us first consider the special case of the cube I^n . We give I the CW structure with two vertices and one edge, so the i^{th} copy of I has a 1-cell e_i and 0-cells 0_i and 1_i , with $de_i = 1_i - 0_i$. The n -cell in the product I^n is $e_1 \times \dots \times e_n$, and we claim that the boundary of this cell is given by the formula

$$(*) \quad d(e_1 \times \dots \times e_n) = \sum_i (-1)^{i+1} e_1 \times \dots \times de_i \times \dots \times e_n$$

This formula is correct modulo the signs of the individual terms $e_1 \times \cdots \times 0_i \times \cdots \times e_n$ and $e_1 \times \cdots \times 1_i \times \cdots \times e_n$ since these are exactly the $(n-1)$ -cells in the boundary sphere ∂I^n of I^n . To obtain the signs in $(*)$, note that switching the two ends of an I factor of I^n produces a reflection of ∂I^n , as does a transposition of two adjacent I factors. Since reflections have degree -1 , this implies that $(*)$ is correct up to an overall sign. This final sign can be determined by looking at any term, say the term $0_1 \times e_2 \times \cdots \times e_n$, which has a minus sign in $(*)$. To check that this is right, consider the n -simplex $[v_0, \dots, v_n]$ with v_0 at the origin and v_k the unit vector along the k^{th} coordinate axis for $k > 0$. This simplex defines the ‘positive’ orientation of I^n as described earlier, and in the usual formula for its boundary the face $[v_0, v_2, \dots, v_n]$, which defines the positive orientation for the face $0_1 \times e_2 \times \cdots \times e_n$ of I^n , has a minus sign.

If we write $I^n = I^i \times I^j$ with $i + j = n$ and we set $e^i = e_1 \times \cdots \times e_i$ and $e^j = e_{i+1} \times \cdots \times e_n$, then the formula $(*)$ becomes $d(e^i \times e^j) = de^i \times e^j + (-1)^i e^i \times de^j$. We will use naturality to reduce the general case of the boundary formula to this special case. When dealing with cellular homology, the maps $f: X \rightarrow Y$ that induce chain maps $f_*: C_*(X) \rightarrow C_*(Y)$ of the cellular chain complexes are the *cellular maps*, taking X^n to Y^n for all n , hence (X^n, X^{n-1}) to (Y^n, Y^{n-1}) . The naturality statement we want is then:

Lemma 3B.2. *For cellular maps $f: X \rightarrow Z$ and $g: Y \rightarrow W$, the cellular chain maps $f_*: C_*(X) \rightarrow C_*(Z)$, $g_*: C_*(Y) \rightarrow C_*(W)$, and $(f \times g)_*: C_*(X \times Y) \rightarrow C_*(Z \times W)$ are related by the formula $(f \times g)_* = f_* \times g_*$.*

Proof: The relation $(f \times g)_* = f_* \times g_*$ means that if $f_*(e_\alpha^i) = \sum_Y m_{\alpha\gamma} e_\gamma^i$ and if $g_*(e_\beta^j) = \sum_\delta n_{\beta\delta} e_\delta^j$, then $(f \times g)_*(e_\alpha^i \times e_\beta^j) = \sum_{\gamma\delta} m_{\alpha\gamma} n_{\beta\delta} (e_\gamma^i \times e_\delta^j)$. The coefficient $m_{\alpha\gamma}$ is the degree of the composition $f_{\alpha\gamma}: S^i \rightarrow X^i/X^{i-1} \rightarrow Z^i/Z^{i-1} \rightarrow S^i$ where the first and third maps are induced by characteristic maps for the cells e_α^i and e_γ^i , and the middle map is induced by the cellular map f . With the natural choices of basepoints in these quotient spaces, $f_{\alpha\gamma}$ is basepoint-preserving. The $n_{\beta\delta}$ ’s are obtained similarly from maps $g_{\beta\delta}: S^j \rightarrow S^j$. For $f \times g$, the map $(f \times g)_{\alpha\beta, \gamma\delta}: S^{i+j} \rightarrow S^{i+j}$ whose degree is the coefficient of $e_\gamma^i \times e_\delta^j$ in $(f \times g)_*(e_\alpha^i \times e_\beta^j)$ is obtained from the product map $f_{\alpha\gamma} \times g_{\beta\delta}: S^i \times S^j \rightarrow S^i \times S^j$ by collapsing the $(i+j-1)$ -skeleton of $S^i \times S^j$ to a point. In other words, $(f \times g)_{\alpha\beta, \gamma\delta}$ is the smash product map $f_{\alpha\gamma} \wedge g_{\beta\delta}$. What we need to show is the formula $\deg(f \wedge g) = \deg(f) \deg(g)$ for basepoint-preserving maps $f: S^i \rightarrow S^i$ and $g: S^j \rightarrow S^j$.

Since $f \wedge g$ is the composition of $f \wedge \mathbb{1}$ and $\mathbb{1} \wedge g$, it suffices to show that $\deg(f \wedge \mathbb{1}) = \deg(f)$ and $\deg(\mathbb{1} \wedge g) = \deg(g)$. We do this by relating smash products to suspension. The smash product $X \wedge S^1$ can be viewed as $X \times I / (X \times \partial I \cup \{x_0\} \times I)$, so it is the reduced suspension ΣX , the quotient of the ordinary suspension SX obtained by collapsing the segment $\{x_0\} \times I$ to a point. If X is a CW complex with x_0 a 0-cell,

the quotient map $SX \rightarrow X \wedge S^1$ induces an isomorphism on homology since it collapses a contractible subcomplex to a point. Taking $X = S^i$, we have the commutative diagram at the right, and from the induced commutative diagram of homology groups H_{i+1} we deduce that Sf and $f \wedge \mathbb{1}$ have the same degree. Since suspension preserves degree by Proposition 2.33, we conclude that $\deg(f \wedge \mathbb{1}) = \deg(f)$. The $\mathbb{1}$ in this formula is the identity map on S^1 , and by iteration we obtain the same result for $\mathbb{1}$ the identity map on S^j since S^j is the smash product of j copies of S^1 . This implies also that $\deg(\mathbb{1} \wedge g) = \deg(g)$ since a permutation of coordinates in S^{i+j} does not affect the degree of maps $S^{i+j} \rightarrow S^{i+j}$. $\square$

$$\begin{array}{ccc} S(S^i) & \xrightarrow{Sf} & S(S^i) \\ \downarrow & & \downarrow \\ S^i \wedge S^1 & \xrightarrow{f \wedge \mathbb{1}} & S^i \wedge S^1 \end{array}$$

Now to finish the proof of the proposition, let $\Phi: I^i \rightarrow X^i$ and $\Psi: I^j \rightarrow Y^j$ be characteristic maps of cells $e_\alpha^i \subset X$ and $e_\beta^j \subset Y$. The restriction of Φ to ∂I^i is the attaching map of e_α^i . We may perform a preliminary homotopy of this attaching map $\partial I^i \rightarrow X^{i-1}$ to make it cellular. There is no need to appeal to the cellular approximation theorem to do this since a direct argument is easy: First deform the attaching map so that it sends all but one face of I^i to a point, which is possible since the union of these faces is contractible, then do a further deformation so that the image point of this union of faces is a 0-cell. A homotopy of the attaching map $\partial I^i \rightarrow X^{i-1}$ does not affect the cellular boundary de_α^i , since de_α^i is determined by the induced map $H_{i-1}(\partial I^i) \rightarrow H_{i-1}(X^{i-1}) \rightarrow H_{i-1}(X^{i-1}, X^{i-2})$. So we may assume Φ is cellular, and likewise Ψ , hence also $\Phi \times \Psi$. The map of cellular chain complexes induced by a cellular map between CW complexes is a chain map, commuting with the cellular boundary maps.

If e^i is the i -cell of I^i and e^j the j -cell of I^j , then $\Phi_*(e^i) = e_\alpha^i$, $\Psi_*(e^j) = e_\beta^j$, and $(\Phi \times \Psi)_*(e^i \times e^j) = e_\alpha^i \times e_\beta^j$, hence

$$\begin{aligned} d(e_\alpha^i \times e_\beta^j) &= d((\Phi \times \Psi)_*(e^i \times e^j)) \\ &= (\Phi \times \Psi)_* d(e^i \times e^j) && \text{since } (\Phi \times \Psi)_* \text{ is a chain map} \\ &= (\Phi \times \Psi)_* (de^i \times e^j + (-1)^i e^i \times de^j) && \text{by the special case} \\ &= \Phi_*(de^i) \times \Psi_*(e^j) + (-1)^i \Phi_*(e^i) \times \Psi_*(de^j) && \text{by the lemma} \\ &= d\Phi_*(e^i) \times \Psi_*(e^j) + (-1)^i \Phi_*(e^i) \times d\Psi_*(e^j) && \text{since } \Phi_* \text{ and } \Psi_* \text{ are} \\ & && \text{chain maps} \\ &= de_\alpha^i \times e_\beta^j + (-1)^i e_\alpha^i \times de_\beta^j \end{aligned}$$

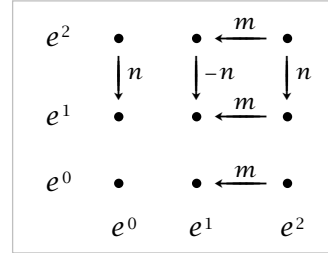
which completes the proof of the proposition. $\square$

Example 3B.3. Consider $X \times S^k$ where we give S^k its usual CW structure with two cells. The boundary formula in $C_*(X \times S^k)$ takes the form $d(a \times b) = da \times b$ since $d = 0$ in $C_*(S^k)$. So the chain complex $C_*(X \times S^k)$ is just the direct sum of two copies of the chain complex $C_*(X)$, one of the copies having its dimension shifted

upward by k . Hence $H_n(X \times S^k; \mathbb{Z}) \approx H_n(X; \mathbb{Z}) \oplus H_{n-k}(X; \mathbb{Z})$ for all n . In particular, we see that all the homology classes in $X \times S^k$ are cross products of homology classes in X and S^k .

Example 3B.4. More subtle things can happen when X and Y both have torsion in their homology. To take the simplest case, let X be S^1 with a cell e^2 attached by a map $S^1 \rightarrow S^1$ of degree m , so $H_1(X; \mathbb{Z}) \approx \mathbb{Z}_m$ and $H_i(X; \mathbb{Z}) = 0$ for $i > 1$. Similarly, let Y be obtained from S^1 by attaching a 2-cell by a map of degree n . Thus X and Y each have CW structures with three cells and so $X \times Y$

has nine cells. These are indicated by the dots in the diagram at the right, with X in the horizontal direction and Y in the vertical direction. The arrows denote the nonzero cellular boundary maps. For example the two arrows leaving the dot in the upper right corner indicate that $\partial(e^2 \times e^2) = m(e^1 \times e^2) + n(e^2 \times e^1)$. Obviously



$H_1(X \times Y; \mathbb{Z})$ is $\mathbb{Z}_m \oplus \mathbb{Z}_n$. In dimension 2, $\text{Ker } \partial$ is generated by $e^1 \times e^1$, and the image of the boundary map from dimension 3 consists of the multiples $(\ell m - kn)(e^1 \times e^1)$. These form a cyclic group generated by $q(e^1 \times e^1)$ where q is the greatest common divisor of m and n , so $H_2(X \times Y; \mathbb{Z}) \approx \mathbb{Z}_q$. In dimension 3 the cycles are the multiples of $(m/q)(e^1 \times e^2) + (n/q)(e^2 \times e^1)$, and the smallest such multiple that is a boundary is $q[(m/q)(e^1 \times e^2) + (n/q)(e^2 \times e^1)] = m(e^1 \times e^2) + n(e^2 \times e^1)$, so $H_3(X \times Y; \mathbb{Z}) \approx \mathbb{Z}_q$. Since X and Y have no homology above dimension 1, this 3-dimensional homology of $X \times Y$ cannot be realized by cross products. As the general theory will show, $H_2(X \times Y; \mathbb{Z})$ is $H_1(X; \mathbb{Z}) \otimes H_1(Y; \mathbb{Z})$ and $H_3(X \times Y; \mathbb{Z})$ is $\text{Tor}(H_1(X; \mathbb{Z}), H_1(Y; \mathbb{Z}))$.

This example generalizes easily to higher dimensions, with $X = S^i \cup e^{i+1}$ and $Y = S^j \cup e^{j+1}$, the attaching maps having degrees m and n , respectively. Essentially the same calculation shows that $X \times Y$ has both H_{i+j} and H_{i+j+1} isomorphic to $\mathbb{Z}_q$.

We should say a few words about why the cross product is independent of CW structures. For this we will need a fact proved in the next chapter in Theorem 4.8, that every map between CW complexes is homotopic to a cellular map. As we mentioned earlier, a cellular map induces a chain map between cellular chain complexes. It is easy to see from the equivalence between cellular and singular homology that the map on cellular homology induced by a cellular map is the same as the map induced on singular homology. Now suppose we have cellular maps $f: X \rightarrow Z$ and $g: Y \rightarrow W$. Then Lemma 3B.2 implies that we have a commutative diagram

$$\begin{array}{ccc} H_i(X; \mathbb{Z}) \times H_j(Y; \mathbb{Z}) & \xrightarrow{\times} & H_{i+j}(X \times Y; \mathbb{Z}) \\ \downarrow f_* \times g_* & & \downarrow (f \times g)_* \\ H_i(Z; \mathbb{Z}) \times H_j(W; \mathbb{Z}) & \xrightarrow{\times} & H_{i+j}(Z \times W; \mathbb{Z}) \end{array}$$

Now take Z and W to be the same spaces as X and Y but with different CW structures, and let f and g be cellular maps homotopic to the identity. The vertical maps in the

diagram are then the identity, and commutativity of the diagram says that the cross products defined using the different CW structures coincide.

Cross product is obviously bilinear, or in other words, distributive. It is not hard to check that it is also associative. What about commutativity? If $T: X \times Y \rightarrow Y \times X$ is transposition of the factors, then we can ask whether $T_*(a \times b)$ equals $b \times a$. The only effect transposing the factors has on the definition of cross product is in the convention for orienting a product $I^i \times I^j$ by taking an ordered basis in the first factor followed by an ordered basis in the second factor. Switching the two factors can be achieved by moving each of the i coordinates of I^i past each of the coordinates of I^j . This is a total of ij transpositions of adjacent coordinates, each realizable by a reflection, so a sign of $(-1)^{ij}$ is introduced. Thus the correct formula is $T_*(a \times b) = (-1)^{ij} b \times a$ for $a \in H_i(X)$ and $b \in H_j(Y)$.

The Algebraic Künneth Formula

By adding together the various cross products we obtain a map

$$\bigoplus_i (H_i(X; \mathbb{Z}) \otimes H_{n-i}(Y; \mathbb{Z})) \longrightarrow H_n(X \times Y; \mathbb{Z})$$

and it is natural to ask whether this is an isomorphism. Example 3B.4 above shows that this is not always the case, though it is true in Example 3B.3. Our main goal in what follows is to show that the map is always injective, and that its cokernel is $\bigoplus_i \text{Tor}(H_i(X; \mathbb{Z}), H_{n-i-1}(Y; \mathbb{Z}))$. More generally, we consider other coefficients besides $\mathbb{Z}$ and show in particular that with field coefficients the map is an isomorphism.

For CW complexes X and Y , the relationship between the cellular chain complexes $C_*(X)$, $C_*(Y)$, and $C_*(X \times Y)$ can be expressed nicely in terms of tensor products. Since the n -cells of $X \times Y$ are the products of i -cells of X with $(n-i)$ -cells of Y , we have $C_n(X \times Y) \approx \bigoplus_i (C_i(X) \otimes C_{n-i}(Y))$, with $e^i \times e^j$ corresponding to $e^i \otimes e^j$. Under this identification the boundary formula of Proposition 3B.1 becomes $d(e^i \otimes e^j) = de^i \otimes e^j + (-1)^i e^i \otimes de^j$. Our task now is purely algebraic, to compute the homology of the chain complex $C_*(X \times Y)$ from the homology of $C_*(X)$ and $C_*(Y)$.

Suppose we are given chain complexes C and C' of abelian groups C_n and C'_n , or more generally R -modules over a commutative ring R . The **tensor product chain complex** $C \otimes_R C'$ is then defined by $(C \otimes_R C')_n = \bigoplus_i (C_i \otimes_R C'_{n-i})$, with boundary maps given by $\partial(c \otimes c') = \partial c \otimes c' + (-1)^i c \otimes \partial c'$ for $c \in C_i$ and $c' \in C'_{n-i}$. The sign $(-1)^i$ guarantees that $\partial^2 = 0$ in $C \otimes_R C'$, since

$$\begin{aligned} \partial^2(c \otimes c') &= \partial(\partial c \otimes c' + (-1)^i c \otimes \partial c') \\ &= \partial^2 c \otimes c' + (-1)^{i-1} \partial c \otimes \partial c' + (-1)^i \partial c \otimes \partial c' + c \otimes \partial^2 c' = 0 \end{aligned}$$

From the boundary formula $\partial(c \otimes c') = \partial c \otimes c' + (-1)^i c \otimes \partial c'$ it follows that the tensor product of cycles is a cycle, and the tensor product of a cycle and a boundary, in either order, is a boundary, just as for the cross product defined earlier. So there is induced a natural map on homology groups $H_i(C) \otimes_R H_{n-i}(C') \rightarrow H_n(C \otimes_R C')$. Summing over i

then gives a map $\bigoplus_i (H_i(C) \otimes_R H_{n-i}(C')) \rightarrow H_n(C \otimes_R C')$. This figures in the following algebraic version of the Künneth formula:

Theorem 3B.5. *If R is a principal ideal domain and the R -modules C_i are free, then for each n there is a natural short exact sequence*

$$0 \rightarrow \bigoplus_i (H_i(C) \otimes_R H_{n-i}(C')) \rightarrow H_n(C \otimes_R C') \rightarrow \bigoplus_i (\text{Tor}_R(H_i(C), H_{n-i-1}(C'))) \rightarrow 0$$

and this sequence splits.

This is a generalization of the universal coefficient theorem for homology, which is the case that C' consists of just the coefficient group G in dimension zero. The proof will also be a natural generalization of the proof of the universal coefficient theorem.

Proof: First we do the special case that the boundary maps in C are all zero, so $H_i(C) = C_i$. In this case $\partial(c \otimes c') = (-1)^i c \otimes \partial c'$ and the chain complex $C \otimes_R C'$ is simply the direct sum of the complexes $C_i \otimes_R C'$, each of which is a direct sum of copies of C' since C_i is free. Hence $H_n(C_i \otimes_R C') \approx C_i \otimes_R H_{n-i}(C') = H_i(C) \otimes_R H_{n-i}(C')$. Summing over i yields an isomorphism $H_n(C \otimes_R C') \approx \bigoplus_i (H_i(C) \otimes_R H_{n-i}(C'))$, which is the statement of the theorem since there are no Tor terms, $H_i(C) = C_i$ being free.

In the general case, let $Z_i \subset C_i$ and $B_i \subset C_i$ denote kernel and image of the boundary homomorphisms for C . These give subchain complexes Z and B of C with trivial boundary maps. We have a short exact sequence of chain complexes $0 \rightarrow Z \rightarrow C \rightarrow B \rightarrow 0$ made up of the short exact sequences $0 \rightarrow Z_i \rightarrow C_i \xrightarrow{\partial} B_{i-1} \rightarrow 0$ each of which splits since B_{i-1} is free, being a submodule of C_{i-1} which is free by assumption. Because of the splitting, when we tensor $0 \rightarrow Z \rightarrow C \rightarrow B \rightarrow 0$ with C' we obtain another short exact sequence of chain complexes, and hence a long exact sequence in homology

$$\cdots \rightarrow H_n(Z \otimes_R C') \rightarrow H_n(C \otimes_R C') \rightarrow H_{n-1}(B \otimes_R C') \rightarrow H_{n-1}(Z \otimes_R C') \rightarrow \cdots$$

where we have $H_{n-1}(B \otimes_R C')$ instead of the expected $H_n(B \otimes_R C')$ since $\partial: C \rightarrow B$ decreases dimension by one. Checking definitions, one sees that the 'boundary' map $H_{n-1}(B \otimes_R C') \rightarrow H_{n-1}(Z \otimes_R C')$ in the preceding long exact sequence is just the map induced by the natural map $B \otimes_R C' \rightarrow Z \otimes_R C'$ coming from the inclusion $B \subset Z$.

Since Z and B are chain complexes with trivial boundary maps, the special case at the beginning of the proof converts the preceding exact sequence into

$$\begin{aligned} \cdots \xrightarrow{i_n} \bigoplus_i (Z_i \otimes_R H_{n-i}(C')) &\rightarrow H_n(C \otimes_R C') \rightarrow \bigoplus_i (B_i \otimes_R H_{n-i-1}(C')) \xrightarrow{i_{n-1}} \\ &\quad \bigoplus_i (Z_i \otimes_R H_{n-i-1}(C')) \rightarrow \cdots \end{aligned}$$

So we have short exact sequences

$$0 \rightarrow \text{Coker } i_n \rightarrow H_n(C \otimes_R C') \rightarrow \text{Ker } i_{n-1} \rightarrow 0$$

where $\text{Coker } i_n = \bigoplus_i (Z_i \otimes_R H_{n-i}(C')) / \text{Im } i_n$, and this equals $\bigoplus_i (H_i(C) \otimes_R H_{n-i}(C'))$ by Lemma 3A.1. It remains to identify $\text{Ker } i_{n-1}$ with $\bigoplus_i \text{Tor}_R(H_i(C), H_{n-i}(C'))$.

By the definition of Tor , tensoring the free resolution $0 \rightarrow B_i \rightarrow Z_i \rightarrow H_i(C) \rightarrow 0$ with $H_{n-i}(C')$ yields an exact sequence

$$0 \rightarrow \text{Tor}_R(H_i(C), H_{n-i}(C')) \rightarrow B_i \otimes_R H_{n-i}(C') \rightarrow Z_i \otimes_R H_{n-i}(C') \rightarrow H_i(C) \otimes_R H_{n-i}(C') \rightarrow 0$$

Hence, summing over i , $\text{Ker } i_n = \bigoplus_i \text{Tor}_R(H_i(C), H_{n-i}(C'))$.

Naturality should be obvious, and we leave it for the reader to fill in the details.

We will show that the short exact sequence in the statement of the theorem splits assuming that both C and C' are free. This suffices for our applications. For the extra argument needed to show splitting when C' is not free, see the exposition in [Hilton & Stambach 1970].

The splitting is via a homomorphism $H_n(C \otimes_R C') \rightarrow \bigoplus_i (H_i(C) \otimes_R H_{n-i}(C'))$ constructed in the following way. As already noted, the sequence $0 \rightarrow Z_i \rightarrow C_i \rightarrow B_{i-1} \rightarrow 0$ splits, so the quotient maps $Z_i \rightarrow H_i(C)$ extend to homomorphisms $C_i \rightarrow H_i(C)$. Similarly we obtain $C'_j \rightarrow H_j(C')$ if C' is free. Viewing the sequences of homology groups $H_i(C)$ and $H_j(C')$ as chain complexes $H(C)$ and $H(C')$ with trivial boundary maps, we thus have chain maps $C \rightarrow H(C)$ and $C' \rightarrow H(C')$, whose tensor product is a chain map $C \otimes_R C' \rightarrow H(C) \otimes_R H(C')$. The induced map on homology for this last chain map is the desired splitting map since the chain complex $H(C) \otimes_R H(C')$ equals its own homology, the boundary maps being trivial. $\square$

The Topological Künneth Formula

Now we can apply the preceding algebra to obtain the topological statement we are looking for:

Theorem 3B.6. *If X and Y are CW complexes and R is a principal ideal domain, then there are natural short exact sequences*

$$0 \rightarrow \bigoplus_i (H_i(X; R) \otimes_R H_{n-i}(Y; R)) \rightarrow H_n(X \times Y; R) \rightarrow \bigoplus_i \text{Tor}_R(H_i(X; R), H_{n-i-1}(Y; R)) \rightarrow 0$$

and these sequences split.

Naturality means that maps $X \rightarrow X'$ and $Y \rightarrow Y'$ induce a map from the short exact sequence for $X \times Y$ to the corresponding short exact sequence for $X' \times Y'$, with commuting squares. The splitting is not natural, however, as an exercise at the end of this section demonstrates.

Proof: When dealing with products of CW complexes there is always the bothersome fact that the compactly generated CW topology may not be the same as the product topology. However, in the present context this is not a real problem. Since the two